

## ORIGINAL RESEARCH OPEN ACCESS

# Agronomic Biofertilizer Potential of a Cyanobacterium-Enriched Culture

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## ABSTRACT

Microalgae- and cyanobacteria-based biofertilizers are valuable resources for sustainable agricultural practices. Though abundant in soils, their biotechnological potential remains underexplored, with only a few species acknowledged for agricultural applications. In this study, the plant growth-promoting effects of a nonaxenic culture, dominated by a filamentous cyanobacterium, were evaluated. In an initial screening, the conditioned culture medium increased root length and stimulated the formation of lateral roots in *Arabidopsis thaliana* and *Lactuca sativa* seedlings, consistent with the cyanobacterium-enriched culture capacity to produce auxins and/or modulate plant auxin metabolism, as observed using auxin-GUS *A. thaliana* reporter plants. Furthermore, cocultivation of the cyanobacterium-enriched culture with *L. sativa* plants, both in pots and more notably in a commercial hydroponic system, resulted in a significant plant growth-promoting effect. In the hydroponic system, cocultivation increased root and shoot biomass by approximately 60%, root length by 30%, and the number of leaves by 20%. This effect was concurrent with enhanced plant carbon and nitrogen metabolism, including higher photosynthesis rate, increased carbohydrate and pigment content, and enhanced activity of nitrogen-related enzymes. Shotgun sequencing led to the identification of a novel cyanobacterial species belonging to the *Tildenella* genus that harbors genomic traits related to nitrogen fixation, siderophore production and auxin biosynthesis. This study contributes to a better understanding of the molecular mechanisms underlying the nonsymbiotic interaction between cyanobacteria and plants and opens new avenues for the use of cyanobacterial species in sustainable hydroponic agricultural practices.

## 1 | Introduction

Biofertilizers based on microalgae and cyanobacteria are considered valuable tools to improve agricultural practices, ensuring food security, and promoting sustainable crop production (Chabili et al. 2024; Jurado-Flores et al. 2025; Kollmen

and Strieth 2022; Makepa et al. 2025; Sun et al. 2026). These microorganisms are primary producers with low nutritional requirements and are ubiquitously distributed across different soil types, from agricultural to arid soils, with positive effects from both ecological and agronomic perspectives (Abinandan et al. 2019). They can complement/replace commonly used

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fertilizers and plant protection products by improving plant nutrition and tolerance to biotic and abiotic stresses (Chabili et al. 2024; Makepa et al. 2025; Poveda 2021; Sun et al. 2026). The application of microbial biomass in soils enhances crop yields by supplying essential nutrients to plants; for example, certain cyanobacteria further contribute through their nitrogen-fixing and phosphate solubilization activities (Chittora et al. 2020). Over the past decades, the cocultivation of several cyanobacterial strains with rice, wheat and other crops has shown promising stimulating effects (Alvarez et al. 2021; Babu et al. 2015; Iniesta-Pallarés et al. 2021). Noteworthy is the effect of the *Azolla-Anabaena* symbiosis on rice production, well-documented all over the world (Bazihizina et al. 2025).

In addition, cyanobacteria represent a vast and largely unexplored source of biologically active compounds, and several have been identified (Abou Jaoudé et al. 2024; Kollmen and Strieth 2022; Sun et al. 2026). It is widely documented that they produce phytohormone-related molecules. For example, the biostimulant properties of *Arthrospira platensis* on the growth of lettuce seedlings are due to cytokinin-like bioactivity (Mógor et al. 2018), while *Oscillatoria angustissima* produces gibberellic acid (Osman et al. 2010), and several cyanobacteria of the genera *Anabaena*, *Nostoc*, *Gleothoece*, and *Plectonema* were reported to be able to synthesize auxin (Duong et al. 2021; Hussain et al. 2010; Mazhar et al. 2013; Tan et al. 2021). Moreover, the polysaccharides secreted by cyanobacteria have biostimulating effects as observed in tomato plants (Rachidi et al. 2020). Furthermore, vitamins, peptides, and phenolic compounds produced by microorganisms are potential improvers of plant growth (Abou Jaoudé et al. 2024; Kollmen and Strieth 2022). Cyanobacteria also contribute to plant defenses by mitigating the effects of biotic and abiotic stresses through the production of biologically active molecules that induce systemic responses (Sarsekeyeva et al. 2024; Kollmen and Strieth 2022; Poveda 2021). For example, the hydrolysate of the cyanobacterium *A. platensis* was able to alleviate the effects of salinity stress on petunia (Bayona-Morcillo et al. 2020), while co-cultivation of lettuce plants with *Oculatella lusitanica* increases plant salinity tolerance (Brito et al. 2022). Furthermore, protection against pathogenic fungi and bacteria was described for crops like tomato, pepper, rice, and wheat treated with different cyanobacterial strains (Gonçalves 2021; Mutale-Joan et al. 2021; Righini et al. 2022; Sun et al. 2026).

Although the effects of cyanobacteria on plants are well documented, the underlying mechanisms remain poorly understood, with only limited insights available (Abou Jaoudé et al. 2024; Chabili et al. 2024; Kapoore et al. 2021; Makepa et al. 2025; Sun et al. 2026). Apart from the effect due to their nitrogen-fixing capacity, several cyanobacteria are known to increase plant nutrition by promoting nitrate assimilation through the stimulation of enzymes like nitrate reductase (NR) and glutamate dehydrogenase (GDH) (Rachidi et al. 2020). Furthermore, an enhancement in the photosynthetic activity and stomatal conductance has been reported following the foliar application of *Anabaena* species and *Microcystis aeruginosa* in willow (Grzesik et al. 2017). Some effects described in different plant species are attributed to the phytohormones secreted by cyanobacteria, which lead to a positive impact on root development, leading to

better plant growth (Mutale-Joan et al. 2020). Through the activation of plant stress responses, such as increased activity of antioxidant enzymes and the accumulation of compounds such as polyphenols and proline, cyanobacteria can also contribute to enhanced tolerance to biotic and abiotic stresses (Poveda 2021). In tomato plants, the injection of polysaccharides from different cyanobacterial strains increased the activity of defensive enzymes like phenylalanine ammonia-lyase (PAL) and antioxidant enzymes, such as aspartate aminotransferase (Farid et al. 2019). In lettuce, the cocultivation with *O. lusitanica* increases proline contents and reduces glutathione, mitigating the deleterious salinity stress effects (Brito et al. 2022), while the relief of drought stress by *Oscillatoria agardhii* in wheat was attributed to the increase in the activity of hydrolytic and antioxidant enzymes, like catalase, superoxide dismutase and peroxidases (Haggag et al. 2018).

This study aimed to evaluate the plant growth-promoting effects of a nonaxenic cyanobacterium-enriched culture. Additionally, an in-depth characterization of the culture was performed using morphological, biochemical, and genomic analyses, showing that the cyanobacterium is a novel species belonging to the *Tildenella* genus. The cyanobacterium-enriched culture had a noticeable plant growth-promoting effect with an impact on lettuce plant physiology, particularly, on photosynthesis and nitrogen use efficiency, as well as enhancing the plant auxin content. The promising results obtained with the agronomic hydroponic system provide a proof of concept for the development of field applications using cyanobacteria-based biofertilizers.

## 2 | Material and Methods

### 2.1 | Sampling, Cyanobacterium Isolation, and Growth

A biological sample (biofilm), mainly comprising a filamentous cyanobacterium, was collected at Quinta do Castro, Anadia, Central Portugal in June 2016 (40°26'37.15"N; 8°26'33.03"W) and subsequently cultured in BG11 medium (Stanier et al. 1971). When visible growth was observed, an aliquot was diluted and streaked onto solid BG11 plates with 1.2% w/v agar (Labkem). Small amounts of biomass were successively re-streaked in new BG11 plates, followed by culture in BG11 liquid medium supplemented with cycloheximide (100 µg mL<sup>-1</sup>), until a unicyanobacterial non-axenic culture was obtained (Dextro et al. 2024; Meixner et al. 2022; Rippka et al. 1979). The cyanobacterium-enriched culture was maintained in BG11 (Roque et al. 2023), in acclimatized chambers under a 16 h light/8 h dark regimen, with continuous agitation at 100 rpm, 22°C, and 40 µmol photons m<sup>-2</sup> s<sup>-1</sup> light irradiance (Fluora, Osram) (subculture A). A second culture (subculture B) was maintained under slightly different conditions (continuous agitation at 70 rpm, 25°C and 15–25 µmol photons m<sup>-2</sup> s<sup>-1</sup>) (MZD LEDtube, Mazda) to promote differential growth of the associated heterotrophic bacteria.

To verify if the cyanobacterial culture has the capacity to fix N<sub>2</sub>, the cells were grown in medium with progressively lower concentrations of NaNO<sub>3</sub> until reaching a medium without nitrate (BG11<sub>0</sub>).

For cocultivation with lettuce plants in pots, the cyanobacterium-enriched culture was grown for 1 month in Erlenmeyer flasks containing perlite that had been washed, sterilized, and soaked in BG11 medium under the same conditions as for subculture A, but without agitation (Brito et al. 2022).

## 2.2 | Cyanobacterium Pigments Quantification

For the quantification of chlorophyll *a*, the procedure provided by Meeks and Castenholz (1978) was followed. Briefly, 2 mL of a 1-month culture was centrifuged, and the pellet was resuspended in 90% (v/v) methanol and incubated for 1 h in the dark with vigorous shaking every 15 min. After final centrifugation, the absorbance of the supernatant was registered at 663 nm, and the chlorophyll *a* content, in  $\mu\text{g mL}^{-1}$ , was determined by multiplying the absorbance by 12.7. For phycobiliproteins extraction, cells were collected by centrifugation at 2500g for 10 min and processed as previously described (Brito et al. 2022). The absorbance spectrum was recorded using a UV-2401PC spectrophotometer (Shimadzu) over 350–750 nm.

## 2.3 | Siderophores and Indole-3-Acetic Acid (IAA) Production by the Cyanobacterial Culture

For siderophore detection, the cells were grown in BG11 or BG11 medium without ferric ammonium citrate for approximately 4 weeks. The chrome azurol S (CAS) assay was performed using 200  $\mu\text{L}$  of 10 $\times$  concentrated conditioned culture medium (1-month-old cyanobacterial culture was centrifuged for 2 min at 10,000g to remove the biomass and obtain the cell-free conditioned medium). The plates were prepared according to Schwyn and Neiland (1987); for details, see also Roque et al. (2023).

Auxin production by the cyanobacterium culture was uncovered by detecting IAA in cell extracts using Salkowski's reagent (Roque et al. 2023) after 3–8 days of growth in BG11 medium or BG11 medium supplemented with 300  $\text{mg mL}^{-1}$  tryptophan under the conditions described above. The cyanobacterial culture extracts were prepared by homogenization of the cells in BG11 solution at 30,000g, using a T10 Basic ULTRA-TURRAX (IKA), followed by centrifugation for 5 min at 10,000g. The quantification of IAA was performed by HPLC-UV/Vis in both cyanobacterial cell extracts and conditioned medium after 3 days of growth in BG11 supplemented with tryptophan (0, 50, 100 and 200  $\text{mg L}^{-1}$ ). The fresh biomass was collected by filtration (filter paper no. 1, Whatman, UK) and prepared following the protocol of Duong et al. (2021) with minor modifications. Approximately 400 mg biomass was extracted with 2 mL of 2-propanol/water/concentrated HCl (2:1:0.002%, v/v/v), and 20  $\mu\text{L}$  of internal standard was added (20  $\text{mg L}^{-1}$  abscisic acid, ABA). The mixture was vortexed and extracted for 60 min in ultrasonics (model 8891, Cole-Parmer) in an ice bath to prevent analyte degradation. After addition of 1 mL dichloromethane, the samples were vortexed, followed by an extraction by ultrasonics for 30 min in an ice bath. The samples were centrifuged at 6000g for 10 min at 4°C (model Z 300 K, Hermle). The organic fraction was collected, evaporated with nitrogen, reconstituted in 1 mL of methanol–water (20:80%, v/v) and filtered using a regenerated cellulose syringe filter (0.45  $\mu\text{m}$ , 15 mm diameter, Phenex,

Phenomenex) for chromatographic analysis. The filtered growth media were directly analyzed after filtering through a regenerated cellulose syringe filter. The extracts were analyzed in an Ultimate 3000 chromatographic system from Dionex (Thermo Fisher Scientific) using a Kinetex C18 (150  $\times$  2.1 mm; 2.6  $\mu\text{m}$ ) column and a precolumn with the same characteristics from Phenomenex and kept at 40°C. The chromatographic separation was performed at 0.3  $\text{mL min}^{-1}$  in gradient mode using (A) methanol (HPLC gradient, Panreac) and (B) ultrapure water (resistivity not less than 18.2  $\text{M}\Omega\text{cm}$ ), both containing 0.1% of formic acid, as follows: 0–12 min, 20%–100% A; 12–15 min, 100% A; 15–17 min, 100%–20% A, and conditions kept for additional 5 min prior to next analysis. The injection volume was 20  $\mu\text{L}$ , and detection was carried out at 280 nm. The quantification of IAA was made by the external calibration method; for biomass internal standard was also used for quantification purposes.

## 2.4 | Plant Material and Growth Conditions

Seeds of *Lactuca sativa* L. cv Marvel of Four Seasons (Vilmorin), *Arabidopsis thaliana*, ecotype Columbia 0 (Col-0) of Nottingham Arabidopsis Stock Centre (NASC, Nottingham, United Kingdom) and transgenic lines transformed with the  $\beta$ -glucuronidase (GUS) reporter gene linked to the DR5 gene promoter (DR5::GUS) (Sabatini et al. 1999) were used.

For the small-scale Petri dish experiments, the seeds were superficially sterilized with 20% (v/v) commercial bleach, washed with sterilized distilled water and germinated in Petri dishes containing Hoagland medium with 1.6% (w/v) bacteriological agar (Labkem) (Roque et al. 2023). The medium was supplemented with 2 mL of cyanobacterial conditioned culture medium (see Section 2.3). For the control groups, 2 mL of BG11 solution was added. Plant growth occurred under controlled conditions: 16 h light (100  $\mu\text{mol m}^{-2}\text{s}^{-1}$ )/8 h dark at 23°C  $\pm$  2°C, 60% rH. Root length and number of lateral roots were evaluated after 10 (*L. sativa*) or 15 (*A. thaliana*) days of growth.

Seeds of *L. sativa* were also germinated in small-scale hydroponic cultures, in pots filled with vermiculite: perlite (2:1) substrate (both products from SIRO, Leal & Soares, S.A.), watered with Hoagland solution, and supplemented, at the top with the perlite in which the organisms were previously grown for 1 month (biomass corresponding to 25  $\mu\text{g}$  of chlorophyll *a* per pot) (Brito et al. 2022). After 4 weeks, the plant biometric parameters (number of leaves, root and shoot length, and plant fresh weight) were assessed. The plant material was preserved using liquid nitrogen and stored at  $-80^{\circ}\text{C}$  for subsequent biochemical analysis.

For large-scale hydroponic cultures, *L. sativa* seeds were grown for 15–21 days in Petri dishes with Hoagland solution supplemented with 1.6% (w/v) agar. The seedlings were detached and transferred to two hydroponic systems, Hydro 60 horizontal (GroHo) (Brito et al. 2022). Ten liters of Hoagland solution were used to fill each growth medium reservoir of the hydroponic systems. In one of the systems, the Hoagland solution was supplemented with the cyanobacterial culture grown for 1 month in the conditions described for subculture A (biomass corresponding to 2.6 mg of chlorophyll *a*). The plants were grown for two

months under environmental conditions in a greenhouse [12 h light ( $25^{\circ}\text{C} \pm 2^{\circ}\text{C}$ )/12 h dark ( $18^{\circ}\text{C} \pm 2^{\circ}\text{C}$ )], and subsequently, the plant biometric parameters (number of leaves, root length, fresh weight and aerial part fresh weight) were evaluated.

## 2.5 | GUS Activity Assay

*A. thaliana* transgenic DR5::GUS seeds were germinated in Petri dishes in the presence of the cyanobacterial culture conditioned medium, as described above. After 10 days of growth, the seedlings were detached, incubated with 90% acetone, washed with sodium phosphate (NaPi) buffer (0.2 M  $\text{NaH}_2\text{PO}_4$ , 0.2 M  $\text{Na}_2\text{HPO}_4$ ), followed by incubation with X-Gluc (5-bromo-4-chloro-3-indolylglucuronide) solution, destained with ethanol and cleared with chloral hydrate as described in Moreira et al. (2024). Finally, the seedlings were observed under a DIC phase contrast microscope Axiolmager AZ (Zeiss) and photographed using an Axiocam MRC3 camera (Zeiss) through the Zen Imaging acquisition software (SP1, Zen 2011).

## 2.6 | Plant Physiological and Biochemical Quantifications

To quantify photosynthesis-related parameters, undamaged leaves were selected, and an infrared gas analyzer (IRGA) LI-6400XT Portable Photosynthesis System (LI-COR) was used to measure  $\text{CO}_2$  assimilation rate, stomatal conductance, transpiration rate, and intracellular  $\text{CO}_2$  concentration, and to determine water use efficiency. The measurements were performed under atmospheric  $\text{CO}_2$  concentration and a saturating photosynthetic photon flux density (PPFD) of  $200 \mu\text{mol m}^{-2} \text{s}^{-1}$ .

For the quantification of plant photosynthetic pigments (chlorophylls *a* [Chl *a*] + *b* [Chl *b*] and carotenoids), 200 mg of material was macerated in 8 mL of 80% (v/v) acetone and centrifuged for 10 min at 4000g, at room temperature, protected from light. The absorbance of the supernatant was measured at 647 nm and 663 nm, and the concentration of the pigments was calculated according to the equations provided by Lichtenthaler and Buschmann (2001). Plant starch and soluble sugars were quantified using anthrone and following the protocol thoroughly described in Roque et al. (2023). Soluble protein content and the activity of the nitrogen metabolism enzymes were evaluated as described earlier by Ferreira et al. (2019); GS (EC 6.3.1.2) was determined by quantification of  $\gamma$ -glutamyl hydroxamate produced by the transferase reaction, and GDH (EC 1.4.1.2) and NR (EC 1.7.1.1) activities were evaluated by spectrophotometrically following NADH consumption.

## 2.7 | RNA Extraction, Reverse Transcription (RT) and qPCR Analysis

Total RNA was isolated from 30 mg of frozen *L. sativa* leaves, as described by Ferreira et al. (2019), evaluated spectrophotometrically for purity and concentration using Multiskan Go Microplate (ThermoFisher Scientific) and assessed for integrity by agarose gel electrophoresis. RT was performed using NZY First-Strand cDNA Synthesis Kit (NZYTech), an oligo(dT)<sub>16</sub>

primer R9:5'-CCAGTGAGCAGAGTGACGAGGACTCGAGCTCAAGCTTTTTTTTTTTT-3', and 1.25  $\mu\text{g}$  of RNA treated with DNase (ThermoFisher Scientific), according to the manufacturer's guidelines.

Quantitative PCR reactions were performed, using CFX96 Real-Time PCR Detection System (BioRad) and PowerUp SYBR Green Master Mix (Applied Biosystems), to quantify the expression of the genes *PSBA* (Gene ID: 3772788) and *RBCL* (Gene ID: 3772894), using specific primers (Table S1), according to the manufacturer. Three technical replicates were performed for each primer pair and sample combination, in 10  $\mu\text{L}$  reaction volume using 400 nM of each primer. The amplification conditions were as follows: UDG activation ( $50^{\circ}\text{C}/2 \text{ min}$ ), initial denaturation ( $95^{\circ}\text{C}/2 \text{ min}$ ) followed by 45 cycles of amplification and denaturation ( $95^{\circ}\text{C}$  for 15 s,  $60^{\circ}\text{C}$  for 60 s) and melting curve generation ( $55^{\circ}\text{C}$  to  $95^{\circ}\text{C}$  read every  $0.5^{\circ}\text{C}$ ). Bio-Rad CFX Manager Software (version 3.1) was used to calculate the cycle threshold ( $C_t$ ) and primer efficiency. To normalize the gene expression data between the different biological samples, the ratio between the  $C_t$  value obtained for each primer pair and the housekeeping genes *TUB2* (Gene ID: 111880334) and *UBC21* (Gene ID: 666261) was calculated.

## 2.8 | Shotgun Sequencing, Genome Assembling, Phylogenetic Analysis, and Functional Characterization

Total gDNA extraction from the cyanobacterium-enriched subcultures (A and B) was performed using the NZY Plant/Fungi gDNA Isolation Kit (NZYTech) according to the manufacturer's instructions. DNA concentration and quality were assessed on 0.8% (w/v) agarose gels, using NanoDrop (Thermo Fisher Scientific) and the Agilent 5400 Fragment Analyzer System (Agilent). The metagenomic shotgun sequencing libraries were prepared and paired-end sequenced ( $2 \times 150 \text{ bp}$ ) at Novogene (Novogene GmbH) using their in-house protocol and the NovaSeq 6000 platform (Illumina). Short-read taxonomic classification was performed using Kaiju v1.9.0 (Menzel et al. 2016) with the NCBI BLAST *nr + euk* database. To generate the assembly, a co-assembly methodology was preferred where the reads from both samples were combined. Co-assembly was performed using MEGAHIT v1.2.9 (Li et al. 2015) and setting a contig minimum length of 1500 bp (--min-contig-len 1500). Three binning tools, MaxBin2 v2.2.7 (Wu et al. 2016), MetaBAT2 v2.15 (Kang et al. 2019) and CONCOCT v1.1.0 (Alneberg et al. 2014) were used and output bins were further dereplicated and aggregated with DASTool v 1.1.7 (Sieber et al. 2018). CheckM v1.2.2 (Parks et al. 2015) lineage workflow (--reduced tree) was used to assess the MAG quality (statistics, completeness and contamination) and the profile utility command to estimate the relative proportion of the MAG within the community (proportion of a bin relative to the number of reads mapped to assembled contigs and adjusted for the size of the bin). Reads were mapped to assemblies using Bowtie2 v2.5.3 (Langmead and Salzberg 2012). The barrnap v0.9 tool (<https://github.com/tseemann/barrnap>) and the CheckM ssu\_finder command were used for rRNA prediction. Prediction of tRNAs was performed with tRNAscan-SE v2.0.12 (Chan et al. 2021). MAG taxonomic classification was performed using the GTDB-Tk v2.3.2 (Chaumeil et al. 2022)

(R08-RS214 release). For additional identity confirmation, the Type (Strain) Genome Server (TYGS; <https://tygs.dsmz.de/>) was used. Phylogenomic analysis was conducted using the genome blast distance phylogeny (GBDP) method implemented at the TYGS platform (Meier-Kolthoff and Goker 2019). A GBDP-based phylogenomic tree was inferred based on the whole proteomes of the selected strains. Pairwise genome comparisons were performed with the closest strain genomes identified to calculate the dDDH and ANI values (determined using the ANIb method implemented in the Pyani algorithm version 0.2.12; Pritchard et al. 2016) and G + C% content difference. A multilocus phylogenomic tree was also constructed using the GToTree v1.8.6 workflow (Lee 2019) with default settings using the single-copy gene set of 251 genes specific for Cyanobacteria and the closely related strains previously identified. Prediction of the putative protein-coding sequences (CDSs) was performed with the Annotate Multiple Microbial Assemblies (v1.073) implemented at the KBase platform (Arkin et al. 2018) that uses the RASTtk toolkit (Brettin et al. 2015). Predicted CDSs were also annotated by the KEGG database (Kanehisa et al. 2023) using BlastKOALA (Kanehisa et al. 2016) with the genus prokaryotes+viruses database. KEGG Mapper was used to assign functional hierarchies and predict KEGG modules. In addition, the genome was screened for the presence of biosynthetic gene clusters (BGCs) with antiSMASH v8.0 (Blin et al. 2025), and when needed, specific annotations were further confirmed by BLASTp searches against the NCBI nonredundant protein database.

## 2.9 | Morphological and Ultrastructural Characterization of *Tildenella* sp.

Cyanobacterium cells were observed under an Axio Lab.A1 light microscope, and micrographs were captured with an Axiocam ERc 5s camera using the ZEN 2.6 software (ZEISS). For the visualization of extracellular polymeric substances (EPS), cells were also stained with Alcian Blue (0.5% w/v in 50% ethanol, v/v) in a 1:1 (culture: dye) ratio (Thornton et al. 2007). For transmission electron microscopy (TEM) analysis, the cyanobacterial culture was centrifuged at 2500g for 10 min, and the collected cells were fixed in 2.5% glutaraldehyde and 2% paraformaldehyde in 0.1 M sodium cacodylate buffer (pH 7.2) (overnight). Cells were then washed three times in sodium cacodylate buffer, followed by post fixation in 2% osmium tetroxide in 0.1 M sodium cacodylate buffer (pH 7.2) (overnight at 4°C), and washed three times in water. Pellets were suspended in 1% (v/v) uranyl acetate (overnight at 4°C) and washed again three times in water. Warm HistoGel (Fisher Scientific) was added to the cell pellets and left to cool at 4°C until solid. Samples were gradually dehydrated with solutions of increasing concentration of ethanol (50%–100%; v/v), and a final step with 100% propylene oxide was repeated three times. Samples were embedded in a series of solutions of EMBed-812 resin (Electron Microscopy Sciences) in propylene oxide of increasing concentrations (25%–100%; v/v) and left in 100% resin for 48 h to fully polymerize. Ultrathin sections were mounted on 300 mesh nickel grids, and samples were stained by the Reynolds' method (Reynolds 1963) for 5 min each. Grids were examined using a JEM-1400Plus electron microscope (JEOL) operating at 80 kV and images were digitally recorded using a CCD digital camera Orious 1100 W (Hamamatsu Photonics).

## 2.10 | Statistical Analysis

Biometric parameters were assessed in at least three independent experiments using 10 (pot trial), 20 (hydroponic trial), and 30 (small-scale experiments) plants. Each biochemical experiment was performed in triplicate, with a pool of at least five plants, and three technical replicates were performed. For the qualification of the IAA, cyanobacterial cultures were grown in triplicate, and for each sample, three technical replicates were prepared.

To compare two datasets, Welch's unequal variances *t*-test was performed. All tests were certified by a 95% confidence level where *p*-value < 0.05, and the results were expressed as mean ± standard error (SEM). The statistical analysis was performed using the GraphPad Prism 6 software (version 6.0.1).

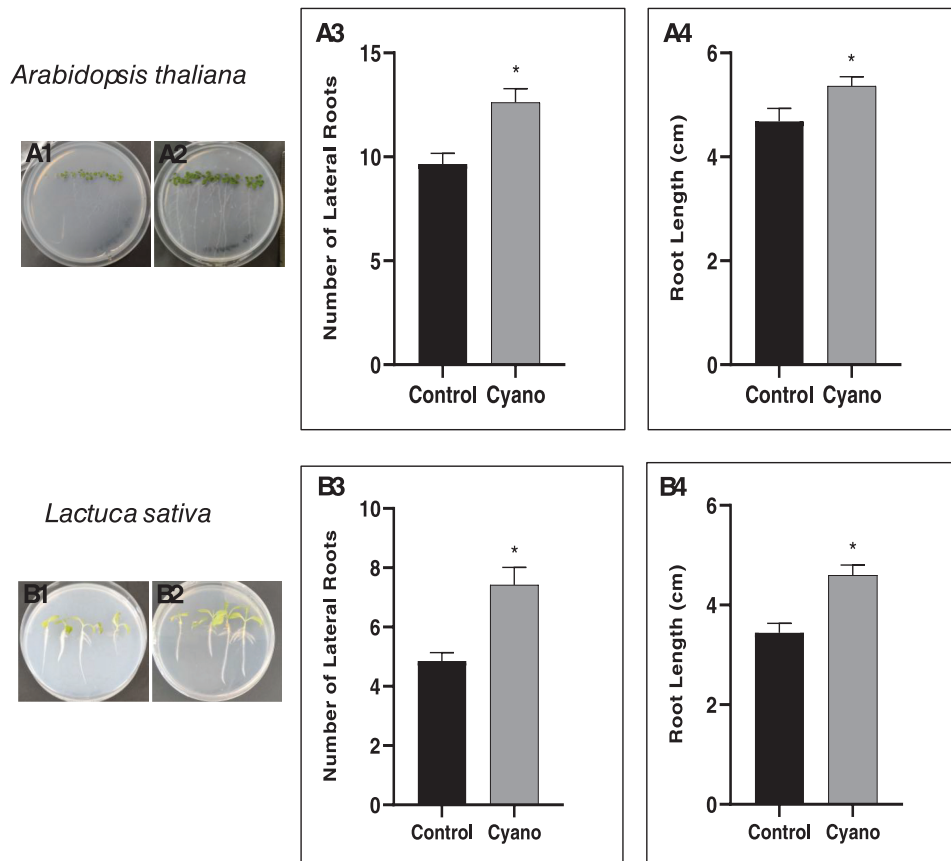
## 3 | Results

### 3.1 | Effect of the Cyanobacterium-Enriched Culture on Plant Growth

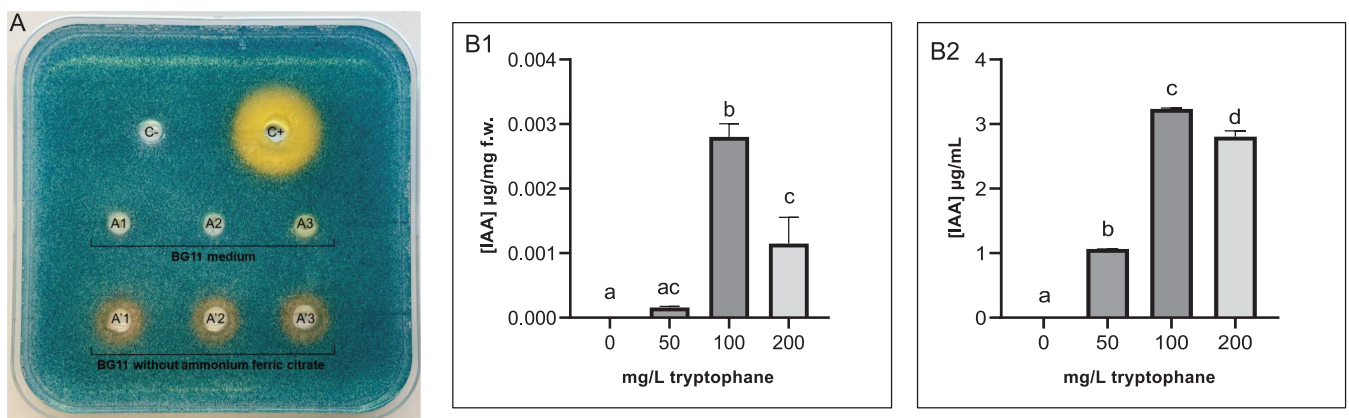
For the isolation of the filamentous cyanobacterial strain, the environmental biofilm was spread in solid BG11 medium, and the resulting culture was subjected to serial dilutions in liquid medium supplemented with cycloheximide to suppress eukaryotic growth until a unicyanobacterial culture was obtained. Since the culture was non-axenic, two subcultures (A and B) were maintained in different conditions to promote differential growth of associated heterotrophic bacteria. To assess whether the cyanobacterial culture had a plant growth-promoting activity, both subcultures were tested by growing *A. thaliana* seedlings in Petri dishes in the presence of conditioned culture medium (cell-free medium) (Figure S1). The results showed that both increase the growth and development of *A. thaliana* plantlets, visibly enhancing root system expansion and shoot development, and stimulating flower production. Since similar effects were observed for both subcultures (Figure S1B,C), the subsequent plant growth-promoting experiments were performed using sub-culture A. To quantify the effects previously observed, seedlings of both the model plant *A. thaliana* and also an agronomic important species, *L. sativa*, were grown in Petri dishes in the presence of conditioned culture medium and the number of lateral roots and root length of the seedlings was evaluated (Figure 1). The results showed a positive effect on root growth and development with increases in the number of lateral roots (30% and 52% in Arabidopsis and lettuce, respectively—Figure 1A3,B3) and in root length (15 and 30%—Figure 1A4,B4).

### 3.2 | Siderophores and Auxin Production by the Cyanobacterium-Enriched Culture

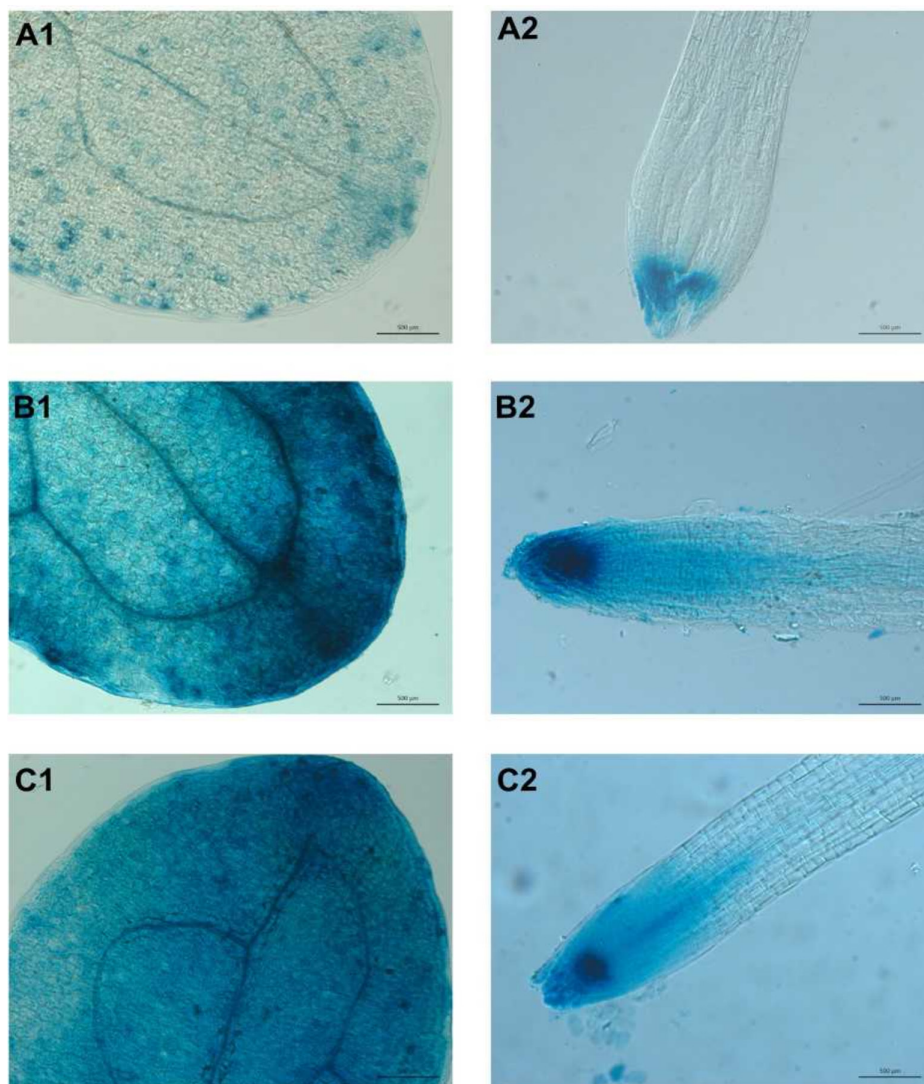
To better understand the mechanisms underlying the beneficial effect of the cyanobacterium-enriched culture on plant growth, we assessed its capacity to release siderophores and to produce the phytohormone IAA. Siderophores' production was assessed using the CAS assay (Figure 2A). In iron-limited conditions (BG11 without ammonium ferric citrate), a yellow halo could be observed around the well containing the concentrated conditioned medium from our culture. This result suggests that our culture produces



**FIGURE 1** | Effect of the cyanobacterium-enriched conditioned culture medium on *Arabidopsis thaliana* (A) and *Lactuca sativa* (B) root growth. *A. thaliana* and *L. sativa* seeds were germinated in Petri dishes with solid Hoagland medium in the absence (A1, B1) or in the presence of the conditioned culture medium (A2, B2) and grown for 10–15 days. The number of lateral roots (A3, B3) and root length (A4, B4) were evaluated in the absence (Control) or in the presence of the conditioned culture medium (Cyano). The results are expressed as means  $\pm$  SEM ( $n = 90$ ). An unpaired *t*-test with Welch's correction, assuming two-tailed *p* values, was performed. \*Statistically significant differences comparing treated plants to the control ( $p < 0.05$ ).



**FIGURE 2** | Detection of siderophores and auxins (indole-3-acetic acid, IAA) production by the cyanobacterium-enriched culture. (A) The detection of siderophores was performed using the Chrome Azurol S (CAS) assay. A1/A'1: *Nostoc* sp. isolate 1.11; A2/A'2: *Nostoc* sp. isolate 2.7 (Biological positive controls, Roque et al. 2023); A3/A'3: Cyanobacterium culture (this work). The microorganisms were grown in BG11 medium or BG11 without ammonium ferric citrate (iron-limited conditions) for approximately 1 month. 200  $\mu\text{L}$  of 10 $\times$  concentrated conditioned medium was placed in each well. C-: Negative control, 10 $\times$  concentrated BG11 medium, C+: Chemical positive control, 100  $\mu\text{L}$  0.5 M EDTA. (B) Detection of IAA quantified by HPLC-UV/Vis in biomass/cell extracts (B1) and conditioned medium (B2) after 3 days of growth in BG11 supplemented with tryptophan (0, 50, 100, and 200  $\text{mg}/\text{L}^{-1}$ ). The results are expressed as means  $\pm$  SEM ( $n = 9$ ). Statistically significant differences within each treatment are represented with different letters (Tukey test,  $p < 0.05$ ).



**FIGURE 3** | GUS staining of leaves and roots of transgenic *Arabidopsis thaliana* plantlets harboring an auxin-sensitive promoter (DR5)-associated beta-glucuronidase (GUS) reporter gene. Plantlets were grown in Petri dishes with a Hoagland medium supplemented with BG11 medium (A1, A2), cyanobacterial conditioned culture medium (B1, B2), or 100 mg/L of tryptophan (C1, C2). Leaves—left panel, roots—right panel.

siderophores, similar to the two *Nostoc* sp. soil isolates previously studied by our research team (Roque et al. 2023).

Initially, the production of IAA by the cyanobacterial culture grown in BG11 supplemented with tryptophan was detected by the Salkowski colorimetric assay (data not shown). Considering these preliminary results, the IAA was then quantified by high-performance liquid chromatography coupled with UV/Vis detection (HPLC-UV/Vis) in cells grown in the absence and presence of tryptophan (50, 100, and 200 mg L<sup>-1</sup>) and respective conditioned medium, after 3 days of cultivation (Figure 2B). The results showed higher IAA production with 100 mg L<sup>-1</sup> of tryptophan in both biomass and conditioned medium.

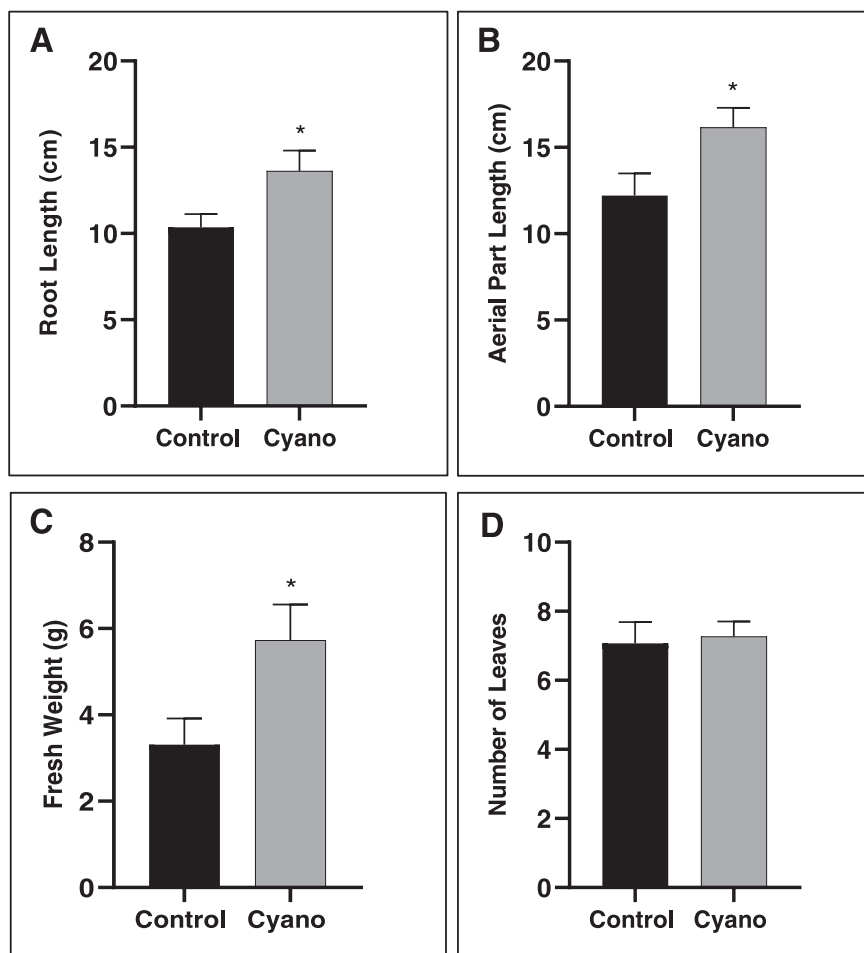
### 3.3 | Cyanobacterium-Enriched Culture Modulates Plant Auxin Metabolism

To evaluate the ability of the cyanobacterium-enriched culture to modulate auxin activity in the plants, Petri dish experiments were settled using *A. thaliana* transgenic seedlings expressing

the GUS reporter gene associated with the auxin-responsive promoter DR5 (DR5::GUS) (Ulmasov et al. 1997) (Figure 3). The results showed an increase in GUS activity in the vasculature and mesophyll cells of leaves and in the apex and vascular central region of roots of the plantlets grown in the presence of conditioned medium (Figure 3B) or in the presence of the auxin precursor tryptophan (100 mg L<sup>-1</sup>) (Figure 3C). As the activity of the DR5 promoter correlates with the auxin content, our results suggest that the cyanobacterium-conditioned culture medium modulates auxin metabolism in *A. thaliana*.

### 3.4 | Plant Growth-Promoting Effect of Cyanobacterium-Enriched Culture

To further demonstrate the plant growth-promoting effect, lettuce plants were cocultivated with the cyanobacterium-enriched culture in a small-scale (Figure 4) and in a large-scale (agronomic scale) hydroponic system (Figure 5). In both cases, increased growth of lettuce plants was evidenced by the increase in root and aerial part length and fresh weight gain. In pots, both



**FIGURE 4** | Effects of the cyanobacterium-enriched culture on *Lactuca sativa* L. growth. Plants were grown in pots in the absence (Control) or in the presence of the cyanobacterial culture (Cyano). Root (A) and aerial part (B) length, total plant fresh weight (C), and number of leaves (D) were evaluated. The results are expressed as means  $\pm$  SEM ( $n = 30$ ). Unpaired Student's *t*-tests with Welch's correction, assuming two-tailed *p*-values, were performed. \*Statistically significant differences ( $p < 0.05$ ).

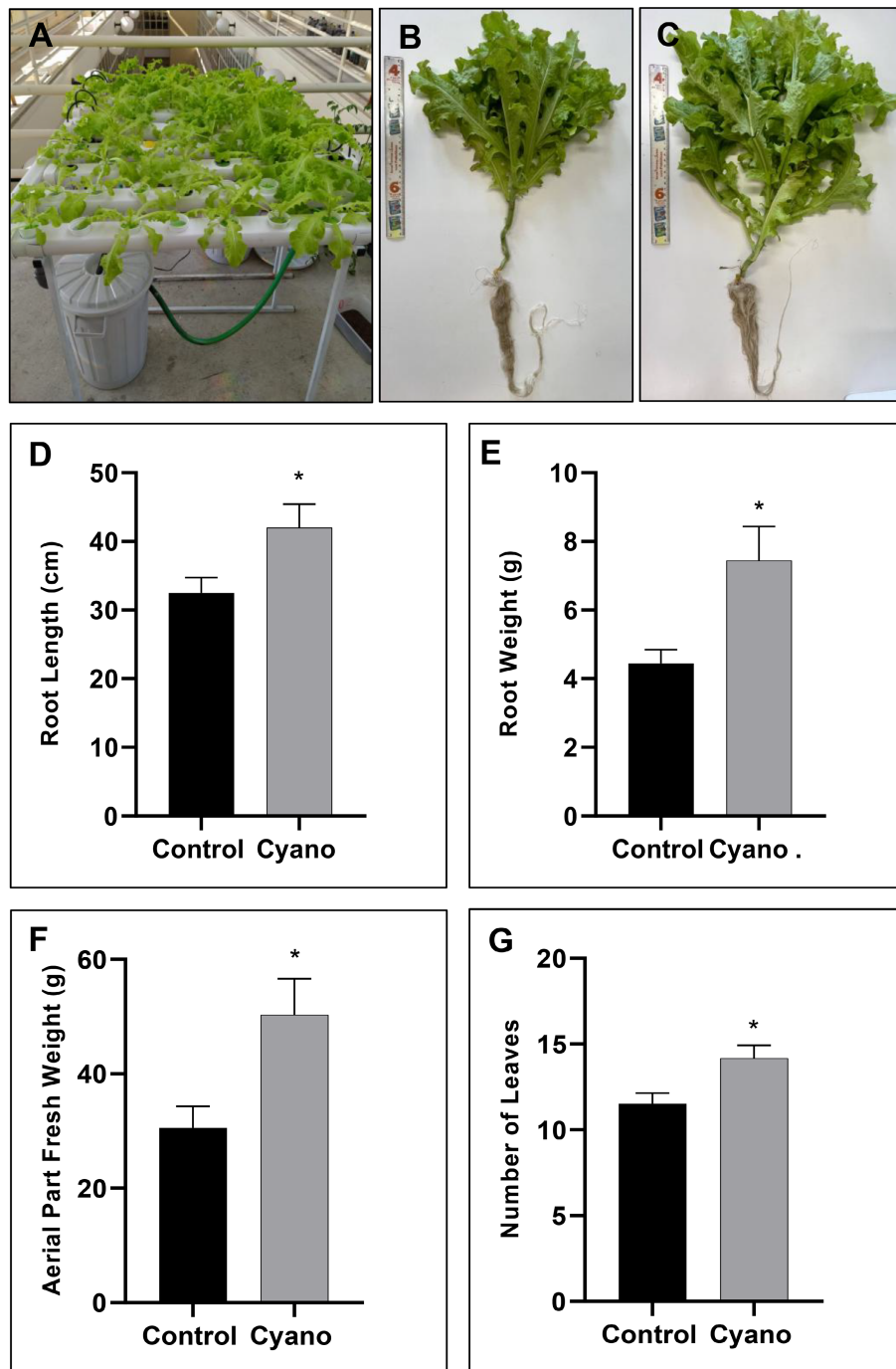
the root and aerial part lengths increased by  $\sim 30\%$  and the total plant fresh weight by  $\sim 70\%$  (Figure 4A–C). In the larger-scale hydroponic system, the culture increased root and shoot biomass by  $\sim 60\%$ , root length by  $\sim 30\%$  (Figure 5D–F), and the growth-promoting effect was also reflected in the number of leaves ( $\sim 20\%$ ) (Figure 5G).

To get an insight into the mechanisms of action behind the plant growth-promoting effect, several biochemical and molecular parameters related to photosynthetic activity/carbon assimilation and nitrogen metabolism were evaluated in the lettuce plants cocultivated with the cyanobacterium-enriched culture in the small-scale hydroponic system (Figure 6). Except for the intracellular concentration of  $\text{CO}_2$  and water use efficiency, all photosynthesis-related parameters increased, such as photosynthetic pigments and carbohydrate content,  $\text{CO}_2$  assimilation, stomatal conductance, and transpiration rate. In addition, the expression of two photosynthesis-related genes, *psbA* (encoding the photosystem II protein D1) and *rbcL* (encoding the ribulose 1,5-bisphosphate carboxylase—Rubisco large subunit), was upregulated. Furthermore, the activity of the enzymes NR and GDH increased, although no differences in the activity of GS were detected, revealing an impact of

co-cultivation on the plant nitrogen metabolism. Altogether, the results showed that the growth-promoting effect exerted by the cyanobacterium-enriched culture is linked to enhanced photosynthetic activity, carbon assimilation, and nitrogen metabolism in lettuce plants.

### 3.5 | Identification of the Cyanobacteria Isolate and Phylogenomic Analysis

To identify the cyanobacterium isolate in the nonaxenic culture, we performed a genome-resolved metagenomics approach to recover the cyanobacterial genome. Samples from the two sub-cultures A and B (see Section 2.1) were sequenced, and after quality control, over 70 million high-quality paired-end reads were obtained for each sample (Table S2). Taxonomic profiling of the short reads revealed that, as expected, the most abundant phylum was Cyanobacteria (47.36% and 40.54% of the classified reads in A and B, respectively; Figure S2). However, over 30% of the reads were assigned to the phylum Pseudomonadota in both samples and 14.27% to the phylum Bacteroidota in sample B. These results corroborate the non-axenic nature of the culture. Although different microbial profiles were obtained,

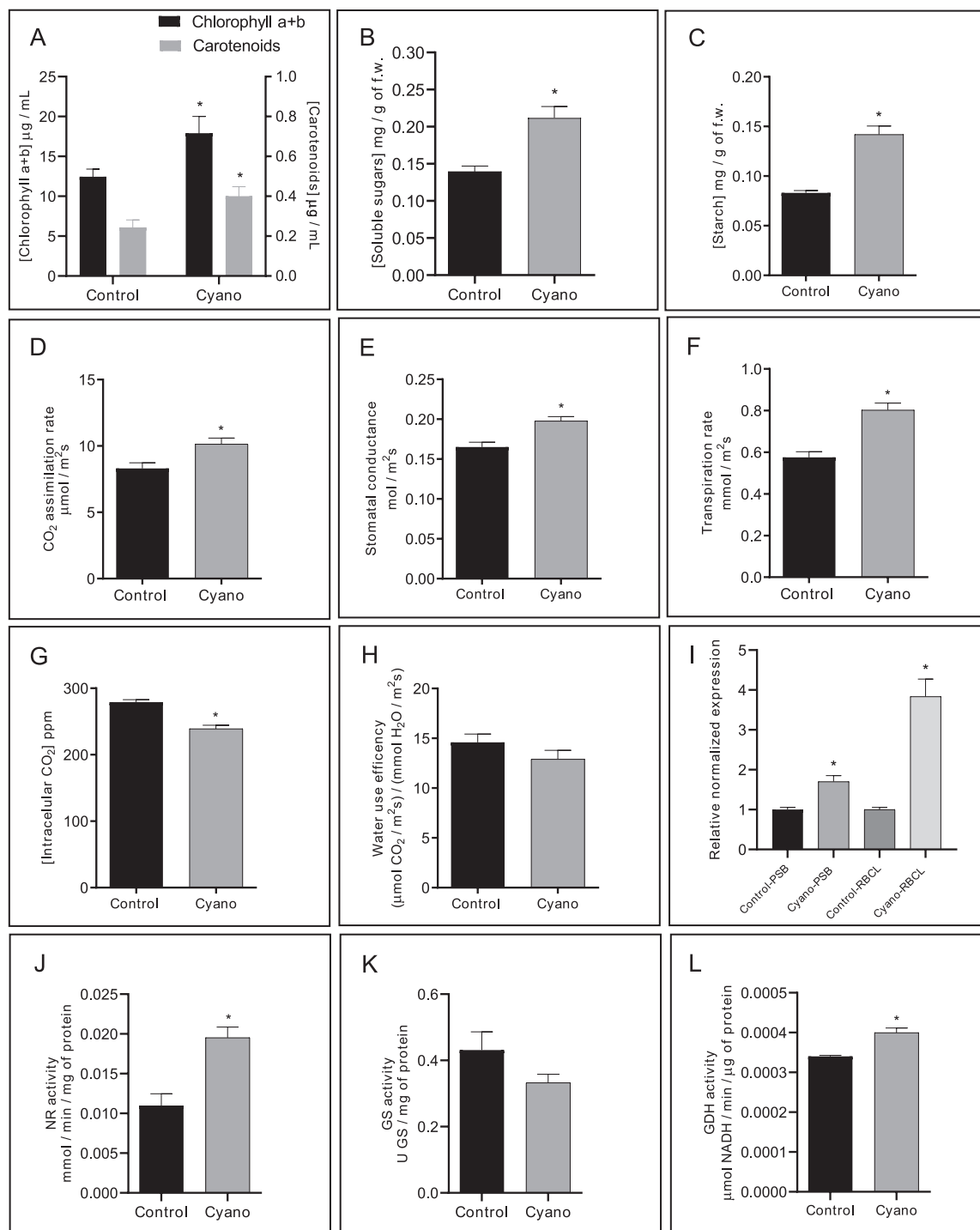


**FIGURE 5** | Effects of the cyanobacterium-enriched culture on *Lactuca sativa* L. grown in a large-scale hydroponic system. Plants grown in an agronomic-scale hydroponic system (A) with Hoagland medium (B) or in Hoagland medium supplemented with the cyanobacterial culture (C). Root length (D), root weight (E), aerial part weight (F), and number of leaves (G) were evaluated in plants grown with Hoagland medium (Control) or in Hoagland medium supplemented with the cyanobacterial culture (Cyano). The results are expressed as means  $\pm$  SEM ( $n=60$ ). Unpaired Student's *t*-tests with Welch's correction, assuming two-tailed *p*-values, were performed. \*Statistically significant differences ( $p < 0.05$ ).

similar growth-promoting effects were observed for both cultures (Figure S1).

Quality-filtered reads were co-assembled, generating an assembly with 28,680 contigs ( $\geq 1500$ bp) and a total assembled length of 229.9 Mb. The contigs were binned into metagenome-assembled genomes (MAGs) using three different binning tools and optimized with the DASTool, generating 26 bins with a score higher than 0.8 (Table S3). Only MAG\_12 was identified in both

samples, with a relative proportion higher than 25%. MAG\_12 scored 99.46% of completeness with 1.81% contamination and was identified as belonging to the Cyanobacteria phylum by the CheckM lineage workflow. A full-length 16S rRNA gene was identified in MAG\_12, and blastn analysis (16S\_ribosomal\_RNA database) showed high sequence identity with the 16S rRNA from *Nodosilinea ramsarensis* strain KH-SS2.6 (98.16% identity; *E*-value 0; Accession number NR\_177034). Taxonomic classification using the GTDB-Tk assigned MAG\_12 to the



**FIGURE 6** | Effect of cyanobacterium-enriched culture on *Lactuca sativa* L. physiology. Plants were grown in pots in the absence (Control) or in the presence of the cyanobacterial culture (Cyano), and several parameters were evaluated. Related to photosynthesis: (A) chlorophyll *a* + *b*, (B) soluble sugar and (C) starch content, (D) CO<sub>2</sub> assimilation rate, (E) stomatal conductance, (F) transpiration rate, (G) intracellular CO<sub>2</sub> concentration, (H) water use efficiency, (I) relative expression of *psbA* and *rbcL* genes determined by qPCR. Related to nitrogen metabolism: activity of nitrate reductase—NR (J), glutamine synthetase—GS (K), and glutamate dehydrogenase—GDH (L). The results are expressed as means ± SEM (*n* = 9). Unpaired Student's *t*-tests with Welch's correction, assuming two-tailed *p* values, were performed. \*Statistically significant differences (*p* < 0.05).

Oculatellaceae family of the Cyanobacteriota phylum. However, no species was assigned as the calculated ANI values with the closest reference strains were below the species delineation cut-off values (95%) established for species delineation (Richter and Rossello-Mora 2009). The genome size of MAG\_12 was 6.59 Mb (6,590,038 bp), with a G + C content of 57.7% (Table 1).

To obtain a better taxonomic resolution, whole genome phylogeny analysis was conducted on MAG\_12 using two methodologies: The GBDP approach implemented at the TYGS Server and a multi-locus phylogenomic analysis using the GTToTree workflow with the single-copy gene set of 251 genes specific for Cyanobacteria. In addition to MAG\_12, the dataset contained 30 genomes (Table S4):

22 genomes were retrieved from the blastn analysis (using the refseq\_genomes database) of the 16S rRNA identified in MAG\_12 that exhibited >97% sequence identity; one genome identified as the closest reference in the GTDB-Tk taxonomic classification (*Tildeniella torsiva* UHER 1998/13D) and 6 genomes retrieved from the TYGS analysis. The genome of *Gloeobacter violaceus* PCC 7421 was used as outgroup (Mai et al. 2018).

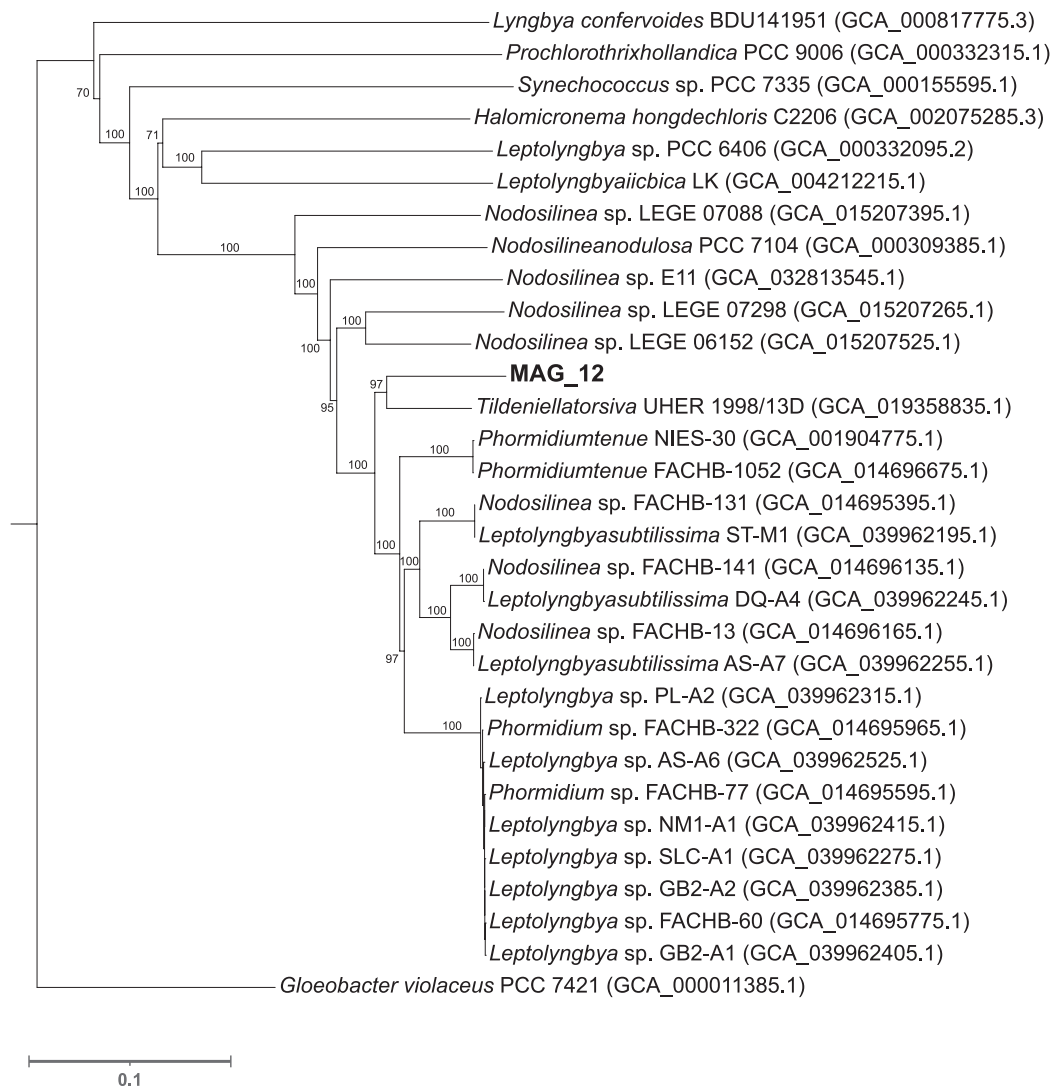
**TABLE 1** | Assembly statistics of *Tildeniella* sp. genome.

|                    |           |
|--------------------|-----------|
| Genome size (bp)   | 6,590,038 |
| No contigs         | 80        |
| N50 (bp)           | 148,191   |
| L50                | 6         |
| G + C content (bp) | 57.70     |
| No CDS             | 6405      |

The GBDP (Figure 7) and the multi-locus phylogenomic trees (Figure S3) were concordant and showed that the closest strain to MAG\_12 was *T. torsiva* UER (Ward et al. 2021), a filamentous cyanobacterium belonging to the Oculatellaceae family (Mai et al. 2018; Strunecky et al. 2023). However, the digital DNA–DNA (dDDH) and ANI values determined in the pairwise comparison (Table S4) retrieved values well below the cut-off values (70% for dDDH and 95%–96% for ANI) to establish new bacterial species (Chun et al. 2018; Richter and Rossello-Mora 2009). These results suggest that MAG\_12 represents a novel cyanobacterial species of the Oculatellaceae family, belonging to the *Tildeniella* genus.

### 3.6 | Functional Genome Annotation

To identify the genomic features that may contribute to the plant growth-promoting effect displayed by the cyanobacterium-enriched culture, we perform the functional annotation of the *Tildeniella* sp. genome. The annotation performed with RAST



**FIGURE 7** | Phylogenomic tree inferred with FastME 2.1.6.1 from whole-proteome-based GBDP distances from MAG\_12 and the selected closely related strains ( $n = 30$ ). *Gloeobacter violaceus* PCC 7421 genome was used as outgroup. The branch lengths are scaled via GBDP distance formula  $d_5$ . Branch values are GBDP pseudo-bootstrap support values > 60% from 100 replications, with an average branch support of 77.7%. The tree was midpoint-rooted and visualized with FigTree v1.4.5. For each strain, the GenBank accession numbers are indicated.

retrieved 6405 CDS, one 16S rRNA and 51 tRNAs (Table S5). The identified CDS were also annotated using the KEGG database, which assigned KO identifiers to 2143 (33.5%) genes that were mapped in 257 KEGG pathways.

The functional analysis revealed several cyanobacteria-specific traits, such as the presence of genes coding for the photosynthetic machinery, including chlorophyll *a*, phycobilins, and the Photosystems I and II. Regarding the phycobiliproteins, the analysis identified the presence of clusters for the biosynthesis of allophycocyanin and phycocyanin. Cell-free extract characterization confirmed the presence of phycocyanin but not phycoerythrin in *Tildeniella* sp. (Figure S4).

Gene clusters associated with nitrogen metabolism were also identified, namely a gene cluster spanning 20 kbp encompassing *nif* genes including the nitrogenase operon (*nifHDK*), the *hox* cluster (13 kbp) containing the genes encoding the bidirectional hydrogenase and genes involved in nitrate assimilation, namely *narB* and *nirA*. In agreement, the experimental results showed that it was possible to grow the cyanobacterial culture in medium without combined nitrogen after a gradual step-down in the nitrate concentration (data not shown).

The most common and well-studied pathways for the biosynthesis of auxin in bacteria involve the conversion of tryptophan (Trp) into IAA via indole-3-pyruvate (the IPA pathway) or via the formation of indole-3-acetamide (the IAM pathway). A third pathway, less studied in bacteria (the IAN pathway), converts Trp to IAA via the formation of indol-3-acetonitrile (IAN) (Lin et al. 2022). Although IAA biosynthesis by the cyanobacterium-enriched culture was detected, no KEGG IAA biosynthetic pathway was mapped in the functional analysis of the *Tildeniella* sp. genome. BLASTp analysis of the deduced proteome of *Tildeniella* sp., using as query the key proteins from the IAA biosynthetic pathway, retrieved MAG12\_2198 (*E*-value 6.03e−34) as a putative indolpyruvate decarboxylase (*ipdC*) responsible for the conversion of IPA to IAA and MAG12\_3284 (*E*-value 0) as a nitrilase responsible for the direct conversion of IAN to IAA (Lin et al. 2022). The BLASTp analysis was extended to the generated MAGs with a relative proportion within the community > 5% (Table S3). The analysis identified the two genes *iaaM* (*E*-value 4.24e−56) and *iaaH* of the IAM pathway in MAG\_4 classified as *Hydrogenophaga intermedia* S1 of the Pseudomonadota phylum. These results need further functional validation to assess their involvement in IAA biosynthesis.

Finally, an antiSMASH analysis was performed to identify potential BGCs for secondary metabolites within the *Tildeniella* sp. genome. The analyses identified six potential BGCs, including three terpenes, one aryl polyene, one triceptide, and one hybrid type I PKS-NRPS cluster. Although all BGCs had no similarities with known compounds, a detailed analysis of the 36 kbp hybrid PKS-NRPS BGC identified the presence of three transport genes: a TonB-dependent siderophore receptor and two genes coding for an iron-siderophore ABC transport system. The presence of these genes suggests that the hybrid PKS-NRPS BGC is potentially responsible for the biosynthesis of a novel siderophore.

### 3.7 | Morphological Characterization of *Tildeniella* sp.

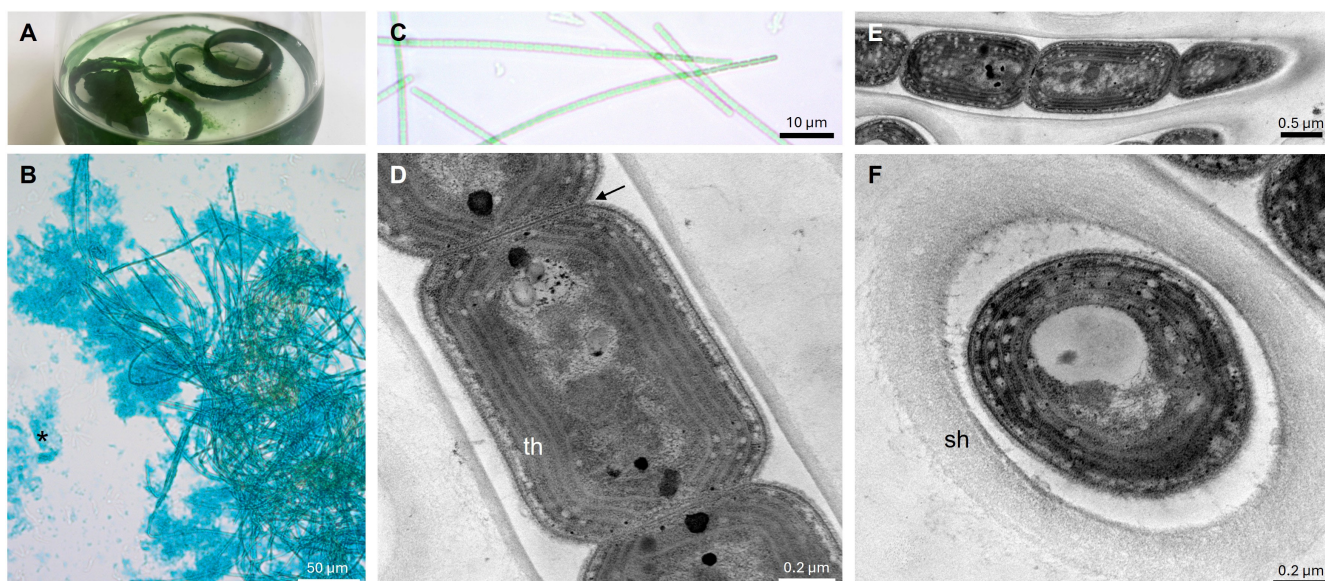
*Tildeniella* sp. cultured in BG11 liquid medium is a filamentous nonheterocystous cyanobacterium that grows as a thick mat with very entangled filaments and a strong production of EPS (Figure 8A,B). Exhibits straight to slightly curved trichomes with cells normally longer than wider, with deeply constricted cross-walls and apical rounded cells (Figure 8C–E). The trichomes are surrounded by a colorless sheath (sh), clearly visible in the TEM micrographs, as well as the parietal arrangement of the thylakoids (th) (Figure 8D,F).

## 4 | Discussion

There is an increasing need to adopt sustainable agricultural practices that could involve the use of cyanobacteria-based biofertilizers. Although plant growth-promoting effects have been clearly demonstrated for certain cyanobacterial strains, the full potential of this beneficial interaction can only be attained once its molecular and biochemical mechanisms are extensively understood (Abou Jaoudé et al. 2024).

In this work, a plant growth-promoting cyanobacterium-enriched culture obtained from an environmental sample was thoroughly characterized. Due to the strong ecological propensity of cyanobacteria to interact with other microorganisms, as well as their production of polysaccharide-rich mucilages that create microenvironments favorable for the growth of associated heterotrophic bacteria, a nonaxenic culture was, therefore, established (Dextro et al. 2024). Shotgun sequencing of two subcultures, grown under different conditions in order to promote the differential growth of the associated heterotrophic bacteria, revealed the prevalence, in both, of a single filamentous cyanobacterium belonging to the Oculatellaceae family and *Tildeniella* genus. This cyanobacterium exhibited similar morphological traits as those previously described for the genus, namely straight to coiled filaments, clearly constricted at the cross-walls, and apical rounded cells (Mai et al. 2018). Associated heterotrophic bacteria were also detected in smaller proportions, with microbial profiles differing between the two subcultures, with only *Tildeniella* sp. being identified in both. This suggests that the observed plant growth-promoting effect can be attributed primarily to the cyanobacterium, although the potential contribution of the remaining heterotrophic bacteria cannot be excluded.

The production of auxins and siderophores has been directly associated with plant growth-promoting effects observed by cyanobacterial cultures (Lin et al. 2022; Lorenzi and Chia 2024). The *Tildeniella* sp.-enriched culture exhibited the capacity to produce both metabolites, and genomic analysis supported its biosynthetic potential. Although the production of IAA by cyanobacteria is well documented, the biosynthetic pathways are poorly characterized, and only genes associated with the IPA pathway were previously identified (Hussain et al. 2015; Lin et al. 2022). Here, we have identified homologues of an indol-3-pyruvate decarboxylase (IPA pathway) and a nitrilase enzyme associated with the IAN pathway in the genome of *Tildeniella* sp. Furthermore,



**FIGURE 8** | Morphology and ultrastructure of the cyanobacterium *Tildenella* sp. The culture forms a thick mat in liquid medium, as it is possible to observe in the Erlenmeyer flask (A), with the filaments intertwined and heavy production of extracellular polymeric substances, as shown with the acidic polysaccharides stain Alcian Blue (B), the release of polysaccharides to the medium is also evident (B, \*). The trichomes are straight to slightly curved (C, E), with constricted cross-walls between the cells (D, arrow). th—thylakoids, sh—sheath.

genes encoding the two enzymes of the IAM auxin biosynthetic pathway were identified in the genome of an associated bacteria (MAG\_4) present in subculture A. On the other hand, siderophores production is well-studied and widely distributed in cyanobacteria (Årstad and Hohmann-Marriott 2019). We have also identified in the cyanobacterial genome a hybrid NRPS-PKS BGC coding for a novel siderophore. These features potentially contribute to the growth-promoting effects observed in *A. thaliana* and *L. sativa*. The cocultivation of the lettuce plants with the cyanobacterium-enriched culture resulted in enhanced growth of both root and aerial parts, highlighting the potential of this culture as a biostimulant in an agronomic context, particularly for the well-established hydroponic cultivation of lettuce.

Cyanobacteria can promote plant growth by regulating key physiological processes, such as photosynthesis (Abou Jaoudé et al. 2024; Gonçalves 2021; Mutale-Joan et al. 2020). Several studies reported the increase in chlorophyll and carbon content in plants biostimulated by cyanobacteria (Gonçalves 2021; Grzesik et al. 2017; Osman et al. 2010; Varalakshmi and Malliga 2012), although direct studies reporting the effects of cyanobacteria on plants' photosynthetic rate are scarce (Grzesik et al. 2017). In this study, the increase in photosynthetic pigments' content, photosynthesis-related parameters, and sugar and starch production indicates a major impact of cocultivation on lettuce plants' carbon metabolism. This was further supported by the increased expression of genes encoding the Rubisco large subunit and the Photosystem II protein D1.

Nitrogen metabolism, essential for plant survival, relies on carbon skeletons from photosynthesis to assimilate ammonium into amino acids. Cyanobacteria have been shown to improve the activity of nitrogen metabolism-related enzymes, such as NR and GDH, in tomato and pea plants (Grzesik et al. 2017; Rachidi et al. 2020). Here, we also observed a significant increase in NR and GDH activity in lettuce plants. In contrast, no change was

observed in GS activity. As GS acts in concert with glutamate synthase (GOGAT) in the primary pathway for ammonium assimilation in plants, this result suggests that nitrogen incorporation via the GS/GOGAT cycle was not significantly affected, but one should keep in mind that GDH can fulfil this role in certain conditions (Moreira et al. 2024). Most frequently, GDH converts glutamate into oxoglutarate needed for the tricarboxylic cycle, but it can also catalyze the reverse reaction in well-documented conditions (Tercé-Laforgue et al. 2015). Although *Tildenella* sp. has the capacity to fix nitrogen, as it grew with  $\text{NaNO}_3$ , the positive effects are unlikely to arise from the transfer of nitrogenous compounds to the plant. Instead, they might be due to the production of several growth-promoting metabolites that can increase the plant's nitrogen use efficiency (Kapoor et al. 2021; Kollmen and Strieth 2022). For instance, cyanobacterial exopolysaccharides (EPS) can stimulate plant nitrogen metabolism and consequently promote plant growth, as previously shown in tomato plants (Rachidi et al. 2020). However, EPS are not the only possible answer for the effect observed, since it is known that cyanobacteria can produce many other bioactive compounds, like vitamins, phenols, terpenoids, free fatty acids, carotenoids, and phytohormones (Chabli et al. 2024; Kapoor et al. 2021; Kollmen and Strieth 2022; Mohamed et al. 2025). By modulating phytohormone levels, cyanobacteria can enhance plant growth by increasing root absorption efficiency, namely through alterations in root architecture (Abou Jaoudé et al. 2024). Here, we observed a positive effect at the root level, with increased elongation, biomass, and lateral root formation, consistent with an auxin-induced response (Woodward and Bartel 2005). Furthermore, we demonstrated that this cyanobacterium-enriched culture has the capacity to produce IAA (HPLC results) and to modulate the plant's auxin content, with an effect similar to that of the auxin precursor tryptophan (GUS activity assay). These results are in line with previous work showing that cyanobacterial strains produce more auxins in coculture with plants, thereby stimulating plant metabolism (Mazhar et al. 2013). Although we

hypothesize that auxin is primarily produced by *Tildeniella* sp., the potential contribution of associated heterotrophic bacteria, namely MAG4 present in one of the subcultures, should not be overlooked. The increased levels of auxin in the plants may also be responsible for the enhanced nitrogen metabolism, photosynthesis rate and carbohydrate contents (Fu et al. 2022; Mazhar et al. 2013). Conversely, nitrogen metabolism can regulate polar auxin transport, increasing root growth (Hu et al. 2020; Ötvös et al. 2021).

The potential of cyanobacteria for sustainable agriculture remains largely unexplored, with few examples of field applications. This study explores the potential application of a novel non-axenic cyanobacterial culture as a biostimulant for sustainable hydroponic agriculture. Furthermore, it sheds some light on the compounds/mechanisms of action underlying the plant responses. Lettuce is widely produced in commercial hydroponic systems, indicating that the methodology used here could be easily scaled up. The cyanobacterium-enriched culture can be cultivated outdoors in bioreactors using a cost-effective mineral medium and sunlight as the energy source and subsequently integrated into established hydroponic systems. Maintaining the culture's long-term viability and purity will be required, and practical considerations (such as the amount of inoculum and its even distribution) must be carefully managed.

## Author Contributions

P.T., P.M., and M.V.M. conceived the research. M.V.M., M.R., A.P., J.P., T.N., C.D., and I.M.V. performed the experiments. M.V.M., P.T., and P.M. analyzed the data and wrote the manuscript. All authors have reviewed and approved the final manuscript.

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## Disclosure

No AI tools were used in the preparation of this manuscript.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The sequencing reads and the *Tildeniella* sp. assembly were deposited in the GenBank repository (BioProject ID PRJNA1295006) under accession numbers SRR34787343, SRR34787344, and SAMN50146540, respectively.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** ppl70807-sup-0001-Supinfo.pdf. **Table S5:** ppl70807-sup-0002-TableS5.xlsx.